

Data Engineering Pipeline for Hotel Booking Analytics Using Pentaho and Power BI

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Abstract— Hotel reservation data frequently contain missing values, duplicate records and inconsistent formats, reducing their effectiveness for business analysis. To address these issues, an end-to-end data engineering pipeline was developed using Pentaho Data Integration to transform raw hotel booking data into a structured warehouse for business intelligence. The ETL process includes data extraction, cleansing, transformation, storage in a MySQL based star schema and visualization through Power BI transformation, storage in a MySQL based star schema and visualization through Power BI. The completed pipeline produced an integrated data warehouse consisting of one fact table and eleven dimension tables while improving data quality through standardization and processing. Interactive dashboard provided insight into different market segments and distribution channels. These results demonstrate that the developed pipeline delivers reliable analytical data supports data driven decision making in the hospitality sector.

Keywords—Data Engineering, Extract, Transform, Load (ETL), Data Warehouse, Star Schema, Pentaho Data Integration, Power BI, Business Intelligence

I. INTRODUCTION (HEADING 1)

The rapid growth of digital service in the hospitality industry has led to a continuous increase in hotel reservation data. Information generated from booking activities, guest profiles, room allocation, and payment transactions provides valuable insight into customer behavior, cancellation patterns and revenue performance. Despite its value, operational data are primarily created for daily transactions rather than analytical purposes. Consequently, missing values, duplicate records and inconsistent format are common, making data less suitable for business analytics and reducing the reliability of data driven decision making [1],[2].

To overcome these challenges, many organizations implement data engineering practices supported by the Extract, Transform and Local (ETL) process and a data warehouse. ETL prepares raw data through extraction, cleansing, transformation and loading before analytical processing begins. This workflow improves data quality while creating a centralized repository for business intelligence applications [1], [3]. Although previous studies have explored ETL, data warehouse and business intelligence individually, only a limited number have demonstrated an integrated pipeline that combines these components for hotel booking analytics [2], [4], [5].

This research develops a data engineering pipeline using Pentaho Data Integration to process the ‘Hotel Booking Dataset’. The processed data are organized into a MySQL

data warehouse using a star schema and visualized through Power BI dashboard. The proposed framework delivers structures and consistent data, enabling more efficient business analysis while illustrating an end-to-end implementation of data engineering in the hospitality domain.

II. RELATED WORK

A. Data Engineering

Data Engineering transforms raw data and reliable information by building and managing the infrastructure required for data processing. It covers data collection, integration, transformation, storage and delivery ensuring that information remains accurate, consistent and accessible for analytical systems and business intelligence applications [4]. Rais and Housley [4] describe data engineering as more than pipeline development emphasizing the importance of maintaining data quality, reliability and scalability throughout the data lifecycle. These principles are applied through an ETL workflow that combines data extraction, Transformation, storage in a data warehouse and visualization with Power BI. The resulting pipeline converts hotel booking records into structure data that can be analyzed efficiently

B. ETL vs ELT

ETL (extract, Transform, Load) and ELT (Extract, Load and Transform) are commonly used methods for integrating data from different sources. The key difference lies in when data transformation takes place. ETL cleans and transforms data before loading it into a data warehouse, while ELT stores raw data first and performs the transformation afterward. The selection of either method depends on the dataset, analytical needs and available computing resources [1], [3]. Structured dataset are generally processes using ETL because data quality issues can be resolved before storage. Handling missing values, duplicate records and inconsistent formats at an early stage improves the reliability of the data used for analysis, ETL is more suitable for cloud environments with scalable storage and processing capabilities [2], [3]. Since the Hotel Booking Dataset is structured, Pentaho Data integration is used to clean, validate, and transform the data before loading it into the data warehouse. This workflow produces consistent data before being ready for business analysis and dashboard visualization.

C. Data Warehouse

A data warehouse integrates data from multiple sources into a centralized repository for analytical processing and decision making. Before storage, data is cleansed, transformed and integrated to improve consistency and create a structure that is more suitable for analysis than an operational database. The design enables faster query execution and supports business intelligence applications through efficient data retrieval and visualization [2],[3]. As a

core component of business intelligence, a data warehouse simplifies data integration and improves the quality of analytical information. Kovalenko et al. [3] highlighted that an effective warehouse design enhances both data integration and decision making capabilities [5]. In this research, MySQL functions as the data warehouse platform, storing processes ETL data before they are connected to Power BI for visualization. The resulting repository provides a structured environment for analyzing hotel booking data efficiently

D. Star Schema

A star schema is a multidimensional data model commonly used in data warehouse design because it simplifies analytical processing. The model organizes into a central fact table linked to multiple dimension tables, making queries and reporting more efficient by reducing table joins [5]. Previous studies have shown that this schema improves query performance and supports multidimensional analysis from perspectives such as time, customer and product [5]. Following this design, the proposed data warehouse organizes hotel booking data into fact and dimension tables, creating a structured model that enables efficient analysis and interactive visualization in Power BI.

E. Pentaho Data Integration

Pentaho Data Integration is an open source ETL platform that simplifies the design of data interaction workflow through a graphical interface. Its built components support extraction, cleansing, validation, transformation and mapping, allowing complex ETL tasks to be completed with minimal coding. These capabilities have made Pentaho a practical solution for data warehouse and business intelligence projects [7].

F. Power BI

Power BI enables processes data to be presented through an interactive dashboard, making business information easier to explore and interpret. By connecting directly multiple data sources, the platform supports data analysis using charts, tables, and key performance indicators (KPIs), allowing users to monitor business performance more effectively [6]. Putri *et al.* [6] reported that Power BI improves information delivery providing interactive visualization and real time reporting. The research is connected to reveal booking patterns, cancellation rates, revenue trends and customer characteristics, helping users gain insight that support business analysis and decision making.

III. SYSTEM DESIGN AND METHODOLOGY

A. Dataset Description

The Hotel Booking dataset, available on Kaggle, is used as the primary key data source in this research. It contains 119,390 reservation records with 36 attributes covering booking for city Hotel and Resort Hotel between July 2015 and August 2017. The data include hotel type, booking dates, length of stay, customer type, distribution channel, booking status and Average Daily rate (ADR). Before entering the ETL pipeline, missing values, duplicate records and formatting inconsistencies are identified and corrected through data cleansing and transformation. These preprocessing steps improve data quality and ensure that the dataset is suitable for data warehouse construction and business intelligence analysis [8].

B. Data Architecture

Four layers from the proposed data architecture include the data source, ETL, data warehouse and visualization. Hotel booking records are imported from a CSV file and processed using Pentaho Data Integration. After Extraction, cleansing, Transformation and loading, processed records are stored in a MySQL based data warehouse. Power BI accesses the components that transform raw operational data into meaningful insight that support business intelligence and data driven decision making.

C. Pipeline Design

The ETL pipeline prepares and validates hotel booking data before storing them in the data warehouse. Records are imported from a CSV file, then cleaned, standardized and transformed through data type conversion, missing value handling, duplicate removal and attribute generation. After preprocessing, the refined data are loaded into a MySQL based data warehouse [1], [7].

D. Tools Selection

The implementation combines Pentaho Data Integration, MySQL, and Power BI to support the complete data engineering workflow. ETL activities are performed in Pentaho before the transformed data are organized in a MySQL star schema. The stored information is then accessed through Power BI to create an interactive dashboard for business analysis and decision making [6], [7].

E. Data Warehouse Design

A star schema organizes the data warehouse into one fact table and eleven dimension tables, separating transactional records from descriptive information to improve analytical performance [5]. The fact table stores booking transactions, while the dimension tables contain detail such as 'hotel', 'guest', 'country', 'arrival_date', 'room_type', 'customer_type', 'market_segment', 'deposit_type', 'distribution_channel', 'agent'. This design reduces join complexity, allowing faster and multidimensional analysis. Consistent relationships between tables are maintained using surrogate keys, while Slowly Changing Dimension (SCD) strategies are applied according to the characteristics of each dimension. These mechanisms preserve data integrity throughout the ETL process and provide a reliable foundation for business analysis and dashboard reporting in Power BI

IV. IMPLEMENTATION

A. Load Raw Data

The first stage of implementation involved moving the Hotel Booking Demand dataset from CSV format into a MySQL staging database. This transformation was built using three main steps in Pentaho Data Integration: a CSV File Input step to read the source file, a Select Values step to pick out the attributes relevant to the ETL process, and a Table Output step to write the results into the raw_booking table. This part of the process went smoothly with no issues encountered. The purpose of this stage is to provide a temporary storage layer that serves as the main reference point for the rest of the ETL pipeline, so that if the pipeline ever needs to be re-run, the raw data remains available

without having to re-read the original CSV file. This transformation flow is shown in Fig. 1.



Fig. 1. Load Raw Data Transformation Using Pentaho Data Integration

B. Data Cleaning

The goal of this stage was to clean and prepare the raw data before it could be used for building the warehouse. A Select Values step was applied first to confirm each attribute carried the correct data type, and to drop the company column, which was removed because over half of its values were NULL, making any standard imputation method such as mean, median, or mode impractical. The agent column had its null values addressed using an If Field Value Is Null step, replacing empty entries with 0 to represent bookings made without a registered agent. A Filter Rows step was put in place to remove records that did not meet basic validity requirements, specifically that adults must be greater than 0, average_daily_rate must be at or above 0, stays_in_week_nights must be greater than 0, and stays_in_weekend_nights must also be greater than 0. Records failing any of these conditions were dropped to prevent distorted results in downstream analysis. A Calculator step was used to generate two new attributes: total_nights, derived from summing stays_in_week_nights and stays_in_weekend_nights, and estimated_revenue, calculated by multiplying average_daily_rate by total_nights to give a rough revenue estimate per booking. A Value Mapper step was applied to group arrival_month into three seasonal categories, where December, January, and February fall under High Season, June, July, and August are labeled Peak Summer, and all remaining months are classified as Regular Season. The cleaned and transformed data was written into the processed_booking table in the processed database layer, keeping it clearly separate from the original raw data. The complete transformation flow is illustrated in Fig. 2.

C. Hotel Dimension

The Hotel Dimension was created to store hotel category information separately from the reservation transaction data. A Table Input step pulls the hotel attribute from the cleansed dataset, a Select Values step narrows it down to the required

Fig. 2. Data Cleansing Transformation Process



columns, and a Unique Rows (HashSet) step removes any duplicates so each hotel type appears exactly once. The deduplicated records are then loaded into the dim_hotel table using an Insert/Update mechanism. No issues were encountered during this stage. Keeping hotel data in its own dimension table supports star schema design by reducing

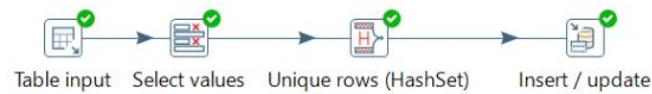


Fig. 3. Hotel Dimension Transformation

redundancy in the fact table and making hotel-based filtering and grouping much more efficient. The transformation flow is presented in Fig. 3.

D. Meal Dimension

The Meal Dimension was built to store information about the meal package types offered across reservations. Data from the cleansed dataset goes through a Select Values step to isolate the relevant attribute, followed by duplicate removal using Unique Rows (HashSet), so that each meal category is represented only once in the table. The result is then loaded into dim_meal using an Insert/Update step, which ensures the table stays consistent whenever new meal types are introduced. No complications arose in this stage. Separating meal data into its own dimension reduces repetition in the fact table and provides a clean reference point for meal-based analysis. The transformation is shown

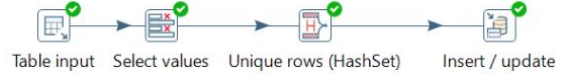


Fig. 4. Meal Dimension Transformation

in Fig. 4

E. Market Segment Dimension

The Market Segment Dimension captures the channel through which each reservation was made, covering categories such as Online Travel Agent, Offline Travel Agent, and Direct Booking. Attribute selection is done through Select Values, followed by deduplication so each segment appears only once before being loaded into dim_market_segment via Insert/Update. Having this data in a dedicated dimension makes it straightforward to compare booking volumes and customer behavior across different acquisition channels. The transformation for this dimension

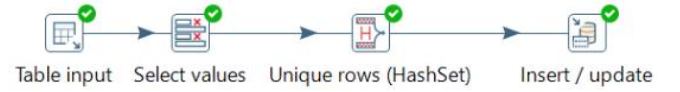


Fig. 5. Market Segment Dimension

is shown in Fig. 5

F. Deposit Type Dimension

The Deposit Type Dimension was built to hold information about the deposit policy applied to each reservation. The processing flow mirrors the other categorical dimensions: attribute selection via Select Values, duplicate removal, and loading into dim_deposit_type through Insert/Update. Keeping deposit type in its own table reduces redundancy and makes it easier to study how different deposit policies relate to booking cancellation

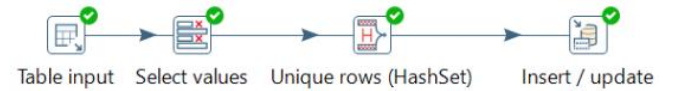


Fig. 6. Deposit Type Dimension Transformation

behavior. The transformation flow is presented in Fig. 6.

G. Customer Type Dimension

The Customer Type Dimension stores classifications of customers based on the nature of their reservation. Data passes through Select Values and Unique Rows (HashSet) before being loaded into dim_customer_type via Insert/Update. The stage completed without any problems.

Isolating customer type into its own dimension keeps the fact table clean and allows for more focused analysis of how different customer segments book and behave over time. The transformation stages are depicted in Fig. 7

H. Date Dimension

The Date Dimension was built to provide a consistent temporal reference for analyzing reservation data. In the

Fig. 7. Customer Type Dimension Transformation

original design, two separate dimensions were planned: dim_date and dim_date_status. After a review session with the supervising lecturer, the decision was made to consolidate both into a single dimension, simplifying the warehouse structure and avoiding unnecessary complexity. The revised transformation selects the required date attributes, removes duplicates, and writes the records into dim_date using a Table Output step. Date data does not change over time, so Slowly Changing Dimension Type 0 was applied here. The dimension supports time-based analysis across granularities such as day, month, year, and



Fig. 8. Date Dimension Transformation

season. The final transformation result is shown in Fig. 8.

I. Guest Dimension

The Guest Dimension stores personal guest information, covering name, email, phone number, credit card, total nights stayed, estimated revenue, and arrival season. Data from the cleansing stage is enriched with additional identity attributes before being loaded into dim_guest using a Dimension Lookup/Update step, which also handles the generation of surrogate keys. The process ran without any issues. Slowly Changing Dimension Type 2 was applied here, meaning any update to a guest's details results in a new record being created rather than the existing one being overwritten. Historical versions of guest data are therefore preserved, which is valuable for tracking behavioral changes over time.



Fig. 9. Guest Dimension Transformation

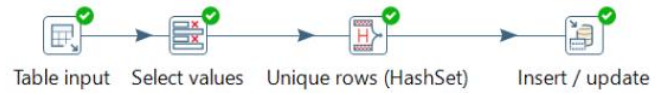
The complete transformation flow is shown in Fig. 9.

J. Distribution Dimension

The Distribution Dimension was built to store information about the channels customers used when making their reservations. The original design for this dimension used Slowly Changing Dimension Type 2, consistent with the approach taken for the Guest Dimension. Upon further review, it became clear that distribution channel data is static in nature and does not carry meaningful historical changes that would need to be tracked. The design was revised to Slowly Changing Dimension Type 0 to reflect this. Attribute selection and deduplication are performed before the data is loaded into dim_distribution, producing a simpler and more maintainable structure. The updated transformation flow is



Fig. 10. Distribution Dimension Transformation



presented in Fig. 10.

K. Room Type Dimension

The Room Type Dimension stores the types of rooms available at the hotel. Like the Distribution Dimension, this one was also initially designed with Slowly Changing Dimension Type 2, and similarly revised to Type 0 once it became clear that room category data does not change over time. The transformation selects the relevant attribute, removes duplicates, and writes records into dim_room_type via Table Output. The structure keeps the fact table free of



Fig. 11. Room Type Dimension Transformation

redundant room data and makes room-based queries straightforward to run. The transformation stages are shown in Fig. 11.

L. Country Dimension

The Country Dimension holds country of origin data for guests making reservations. One additional step sets this dimension apart from the others: a Filter Rows step is included after deduplication to catch and remove any remaining NULL values in the country attribute. Valid records are then loaded into dim_country using Insert/Update. Country-level data enables geographic breakdowns of reservation volume, making it possible to identify which

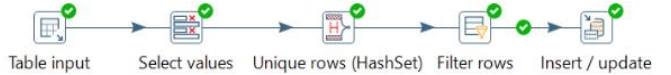


Fig. 12. Country Dimension Transformation

regions contribute the most bookings or exhibit distinct patterns. The transformation flow is depicted in Fig. 12.

M. Fact Booking

Fact Booking serves as the central table in the data warehouse, holding the full set of reservation transaction records. The transformation retrieves the cleansed data and performs lookups against all ten dimension tables to resolve surrogate keys for hotel, meal, market segment, deposit type, customer type, date, guest, distribution, room type, and country. Ten lookups running per row across a large dataset meant this stage took noticeably longer to complete than any of the dimension transformations, which was expected given the volume of operations involved. Once all keys were resolved, the records were written into fact_booking. The resulting table ties every dimension together into a single star schema, making it possible to analyze reservations from virtually any angle, whether that is cancellation rates, revenue patterns, seasonal trends, or guest demographics.

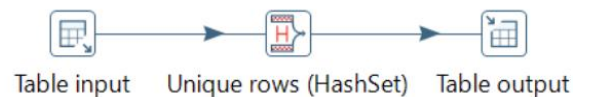


Fig. 13. Fact Booking Transformation

The transformation is illustrated in Fig. 13.

N. ETL Job (Orchestration)

An ETL Job was configured in Pentaho Data Integration to manage the execution order of every transformation in the

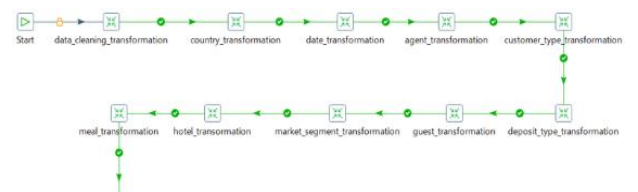


Fig. 14. ETL Job Transformation Using Pentaho Data Integration

pipeline. The sequence starts with Load Raw Data, moves through Data Cleansing, runs each dimension transformation, and ends with the Fact Booking load. The order is intentional: dimension tables must be fully populated before the fact table can perform its lookups. Running everything through a single orchestrated job eliminates the need to trigger each transformation manually and reduces the risk of steps being skipped or executed out of order. The complete orchestration flow is shown in Fig. 14.

V. RESULT AND DISCUSSION

Using Pentaho Data Integration tools, an ETL pipeline was successfully constructed for this project. A structured data warehouse prepared for business analysis was produced by the pipeline. Following the star schema model, it generated eleven dimension tables and one fact table. These tables comprise the booking fact table and the following dimension tables: hotel, date, guest, meal, country, market segment, distribution channel, room type, deposit type, agent, and customer type. All tables were successfully loaded into a MySQL database named "hotel_mart_data_warehouse." Furthermore, the resulting data warehouse provides a reliable foundation for data visualization and data-driven decision-making processes.

In addition to creating a data pipeline, this project developed a visualization dashboard to support analytical needs. Visualizations based on hotel booking data from 2015 to 2017 reveal a total of 6.51 million bookings with a cancellation rate of 39%. During this period, the hotel generated \$3.23 billion in total revenue, with an average daily rate of \$103.29. The dashboard includes a dedicated page for analyzing cancellation trends; this analysis shows that bookings made more than 180 days in advance without a deposit have a very high cancellation rate of 49%. Conversely, bookings made with a lead time of zero to seven days show a cancellation rate of only 12%. Furthermore, 89% of all reservations were made without a deposit, which is in line with the distribution of deposit types. The "no deposit" category has a greater cancellation rate than the "non-refundable" category, according to data. Additionally, guests who have canceled once or twice in the past have a 95% chance of canceling future reservations, according to a review of historical cancellation data.

Naturally, total revenue performance is greatly impacted by a high cancellation rate. As a result, revenue analysis was carried out from a number of angles. In August, for example, hotel income peaked at \$420 million for city hotels and \$340 million for resort hotels. This is consistent with the city hotels' Average Daily Rate (ADR) of \$67 million for the same month. Additionally, the visualization divides the months into seasons according to the number of visitors; the study shows that the normal season brought in \$1.41 billion, while the peak summer period brought in \$1.47 billion for both hotel kinds. This pattern demonstrates that demand is highest in July and August, especially for resort hotels, which are mostly driven by leisure travel.

The traits of various consumer categories are intimately related to the significant linkage between revenue patterns and particular timeframes. With 3.27 million reservations, Online Travel Agencies (OTAs) dominated market categories; nevertheless, their success rate was only 58.61%,

and their cancellation rate was 41%. On the other hand, while having far smaller booking volumes, the Corporate and Direct segments performed better, with success rates exceeding 82% and cancellation rates of 17% and 18%, respectively. The Groups category, on the other hand, had the highest cancellation rate (60%) and the lowest success rate (39.92%), which was probably caused by the longest average lead time (179.81 days). A distribution channel analysis was also carried out, and Travel Agencies/Tour Operators (TA/TO) was shown to have the highest booking volume by a considerable margin, considerably outperforming Direct, Corporate, GDS, and Undefined channels. Furthermore, the bulk of visitors belonged to the transient category, which dominated practically every market segment. The guests' geographic distribution showed a wide worldwide reach, with the biggest concentration coming from Europe.

Based on the analysis from the visualization dashboard, several business strategies are recommended for hotel management to implement. To mitigate high cancellation rates, management could introduce a deposit policy for bookings, particularly those made more than 90 days in advance, such as requiring a partial upfront payment or offering discounts for full prepayment. Additionally, a non-refundable policy could be applied to guests with a history of frequent cancellations to reduce booking cancellation risks. By increasing the Average Daily Rate (ADR) during the busiest summer months, the hotel could maximize its dynamic pricing approach and increase income. Additionally, a market segment analysis indicates that management should devote resources to fortifying ties with the Corporate and Direct segments; this could be accomplished by putting in place loyalty programs and competitive corporate offers to boost the proportion of reservations from these groups.

VI. CONCLUSION

Using the Hotel Booking Demand dataset, this project effectively created a thorough data engineering pipeline for the hotel industry. The pipeline leverages Power BI for data visualization, MySQL as a data warehouse with a star schema, and an ETL architecture created with Pentaho. In order to transform raw, inconsistent, and duplicate data into a clean, analysis-ready data warehouse, data transformation procedures such as validation, management of missing information, filtering, deduplication, and the development of new attributes proved successful. Eleven dimension tables and one fact table were successfully created as a result of the process.

Booking time, deposit type, and cancellation history all had a major impact on the 39% cancellation rate out of 6.51 million total bookings between 2015 and 2017, according to analysis done using the Power BI dashboard. August had the largest earnings during the summer, when revenue peaked. Additionally, even while Online Travel Agencies (OTAs) and TA/TO distribution channels continued to dominate booking volumes, the Corporate and Direct segments showed the best booking quality. This project shows how a well-thought-out ETL pipeline can produce a technically sound data architecture and useful business insights to help the hotel sector make data-driven decisions.

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